Microbiology 2

Science

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South East Technological University

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Microbiology 2 (A36021)

Short Title: Microbiology 2
Faculty Science and Computing
Department Science
Cluster Biology
Author NKENNEDY
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module is intended to build on the microbiology knowledge and skills introduced in the module 'Microbiology 1'. It explores the evolution and diversity of microbes, including bacteria, archaea, fungi and viruses.

Programmes

`SCIE-0045`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	3 / 5 / M

Indicative Content

- Evolution and differences/similarities between the three domains of life, with a focus on microbial branches (bacteria/archaea/fungi)
- Viruses, including bacteriophage life cycles
- Genotypic and phenotypic methods for microbial classification
- Microbial metabolism
- Cultivation of micro-organisms: culture maintenance, media, equipment, sterilisation
- Identification and enumeration of micro-organisms
- Micro-organisms and disease
- Practicals: isolation & identification of a range of microorganisms, using different culturing methods

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Complete the isolation, classification, and maintenance of a wide range of micro-organisms.
2. Prepare media for culture of micro-organisms.
3. Compare the methods used in microbial taxonomy, both phenotypic and genotypic.
4. Differentiate between major groups of bacteria, fungi and viruses important in molecular and biopharmaceutical science.
5. Analyse the mechanisms by which micro-organisms cause disease.
6. Interpret current knowledge and scientific research related to a particular microorganism.

Learning and Teaching Methods

- Lectures on the indicative content will be delivered, incorporating active learning activities and formative assessment (ungraded quizzes).
- Laboratory investigations will reinforce and support theoretical elements of the module.
- In-class presentation on a microbial species of the student's choice (peer-to-peer learning).

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	1,2,3
Presentation	20%	6
In-Class Assessment	30%	3,4,5

Assessment Criteria

<40%: No understanding of the basic principles of microbiology and the handling of micro-organisms.

40%-49%: The student has a basic understanding of the fundamental principles of the module. May have some difficulty with applying this theory.

50%-59%: The student has a good understanding of microbiology both theoretical and the evaluation of practical data.

60%-69%: In addition to the above shows a high level of competence and efficiency in microbiology.

70%-100%: Student demonstrates a high level of understanding of microbiology plus is able to critically evaluate and discuss the significance of the module.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	36	
Independent Learning	75	

Essential Material

- "Microbiology 2 module Moodle page." http://moodle.wit.ie
- Madigan, M.T., J.M. Martinko, D.A. Stahl and D.P. Clark and Brock, T. _Brock Biology of Microorganisms_. 13th ed.. UK: Benjamin Cummings (Pearson), 2011.

Supplementary Material

- Collins, C., J. Grange, P. Lyne and J. Falkingham. _Collins and Lyne's Microbiological Methods_. 8th Edition. London: Arnold, 2003.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab